

Mann Janodia

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Professional profile

MSc Robotics Engineer specializing in AI-driven feedback loops and safe human-robot interaction in confined spaces. Leveraging a background in ISRO hardware verification and Bosch Rexroth automation, I develop predictable, real-world systems for 6-DOF manipulators. I focus on bridging the gap between simulation and physical deployment through rigorous experimental research, data collection, and technical reporting. I have Knowledge of Mechanical Designing in software as well as using machineries in real life.

Experience

Radio Frequency Engineer Intern | Optimized Solutions Limited (January 2025 – April 2025)

Designed and tested RF-enabled embedded hardware systems from prototype through validation.

Conducted signal analysis using oscilloscopes and spectrum analysers to evaluate system performance.

Performed environmental testing across temperature ranges up to $\pm 85^{\circ}\text{C}$ to verify hardware reliability.

Diagnosed and resolved recurring hardware faults through schematic interpretation and systematic debugging.

Produced technical documentation describing testing procedures, results, and compliance checks.

Electrical Engineering Intern | Bosch Rexroth | India (April 2024 – May 2024)

Tested hydraulic and electromechanical systems including pumps, industrial motors, and sensor assemblies.

Conducted system fault diagnosis and sensor calibration during validation procedures.

Worked with electrical, mechanical, and hydraulic engineering teams during equipment testing and system verification.

Firmware Lead | Robofest 4.0 | India 2024

Led firmware development and hardware/electronics integration for a competitive robotics platform, responsible for embedded software architecture, sensor interfacing, and motor control pipelines.

Secured ₹50,000 innovation grant; patent filed for embedded control system.

Projects

Niryo NED2 Test Tube Sorting System

Built a fully autonomous 3-component lab system: TurtleBot4 (mobile navigation), Niryo NED2 (6-DOF manipulation), and ESP32+PN532 RFID reader — integrating hardware, software, and communications under close academic supervision.

Camera Servo Tracking — Raspberry Pi 4

Deployed YOLOv8 real-time object detection at 30fps on Raspberry Pi 4; designed proportional control loop for MG996R pan-tilt servo system via GPIO PWM — hands-on embedded electronics.

Autonomous Mapping Robot

Designed and built the complete physical robot structure in SolidWorks — chassis, sensor mounts, motor housings, and wheel assembly — then integrated with dual-Arduino electronics and firmware.

Industrial Robotic Arm Simulation | ABB Studio

Designed and built the Programmed 6-DOF arm for pick-and-place across 4+ operating scenarios with sensor processing and network integration.

Self-Balancing Robot | Robofest 4.0

Custom PID control loop using IMU 6050 data; 30% reduction in oscillation through Kp, Ki, Kd tuning. Secured ₹50,000 grant and progressed to patent filing.

IoT Sensor Network & Predictive Maintenance

ESP32 real-time hydraulic pressure monitoring (sub-500ms latency); Python/Scikit-learn motor failure model at 85%+ accuracy across 10,000+ samples.

Skills

Mechanical CAD & Design: SolidWorks (3D modelling, assemblies, engineering drawings), Fusion 360, AutoCAD, PTC Creo (conceptual familiarity), tolerance analysis.

Electronics & Hardware Testing: Oscilloscopes, spectrum analysers, digital multimeters, RF signal analysis, PCB-level fault diagnosis, hardware validation, environmental testing ($\pm 85^{\circ}\text{C}$).

Embedded Systems: ESP32, Raspberry Pi 4, Arduino Uno, sensor integration (RFID, IMU, ultrasonic, servo), pigpio, motor control.

Robotics & ROS2: ROS2 Jazzy, Nav2, SLAM, MoveIt2, PyNiryo, URDF/Xacro, RViz2, 6-DOF manipulation, autonomous navigation.

AI & Computer Vision: YOLOv8 NCNN, MediaPipe, OpenCV, custom NN from scratch (NumPy), TensorFlow/Keras, Scikit-learn.

Programming: Python (advanced), C/C++ (intermediate), Bash, YAML, Git, Docker, Linux (Ubuntu 24.04), Flask REST APIs.

Simulation & Design: Webots, Gazebo, ABB RobotStudio, SolidWorks, AutoCAD, Fusion 360.

Education and qualifications

MSc Robotics Engineering | Middlesex University

(2025 - 2026)

BEng Mechatronics Engineering (Minor Degree in AI and ML) | GCET

(2021 – 2025)

Minor Modules (AI/ML): Artificial Intelligence, Machine Learning Fundamentals, Deep Learning, Algorithm Design, Control Systems

Achievements

Patent under process - Self-Balancing Robot (embedded control & hardware stability)

ISRO Training Programme - hardware verification for extreme-environment space applications

Robofest 4.0 Grant Recipient - ₹50,000 for hardware innovation in autonomous robotics

Live Portfolio - 5 deployed robot projects at mannjanodia3404.github.io - resume downloadable directly from portfolio.